



Sen2Fire

Data Analytics





Introduction

Sen2Fire: a challenging benchmark dataset for wildfire detection using sentinel data

26 Mar 2024 by Yonghao Xu et. al.

Task: wildfire detection via binary semantic segmentation: develop a deep learning model to classify each pixel in a 13-channel satellite image as either '*fire*' or '*non-fire*'.





Dataset

Dataset



Dataset Information

- No geographic overlap between the training, validation and test sets.
- Each patch has a size of 512×512 pixels.
- 13 channels
 - 12 multi-spectral bands from Sentinel-2.
 - 1 aerosol index product channel from Sentinel-5P.
- All data is resampled to a consistent 10-meter spatial resolution.

2466

Patches

.npz

Format



Dataset

Dataset



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Scene 1

Training Set, 864 patches.



Dataset

Dataset



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Scene 2

Training Set, 594 patches.



Dataset

Dataset



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Scene 3

Validation Set, 504 patches.



Dataset

Dataset



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- All data is resampled to a consistent 10-meter spatial resolution.

Scene 4

Test Set, 504 patches.



Dataset

Ratio of the fire samples to the non-fire samples

Training Set



Validation Set



Test Set

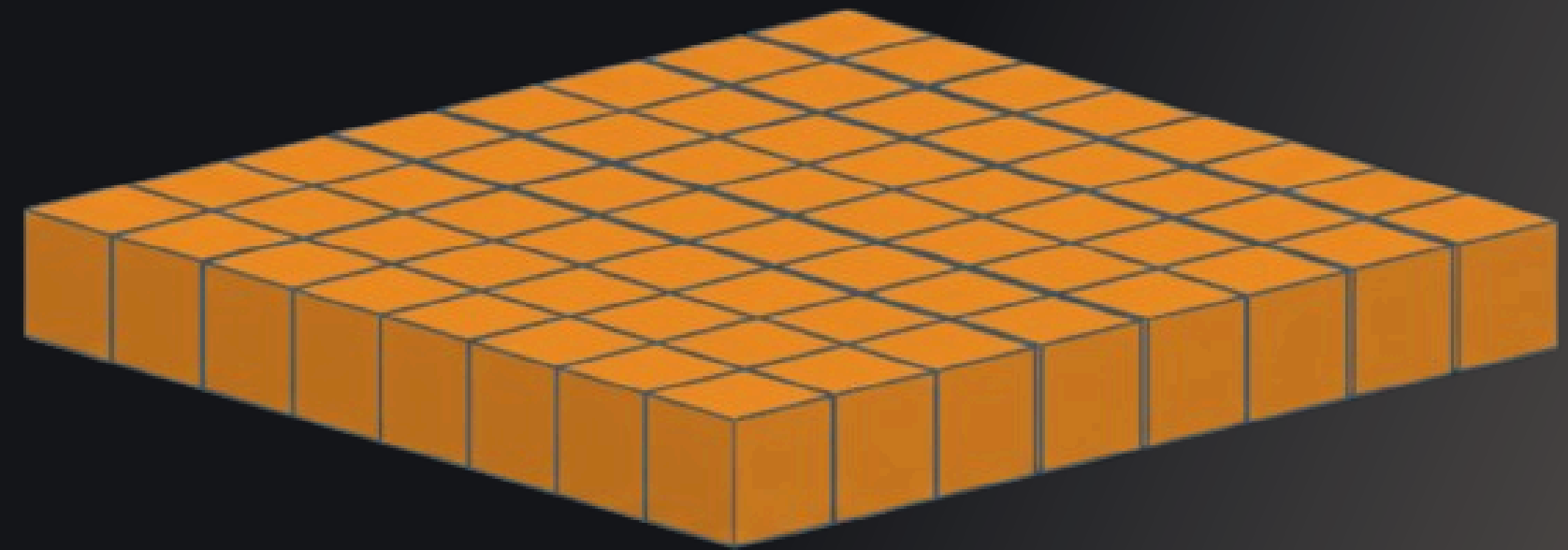




Anatomy of a Data Patch

Layer 1 - Coastal Aerosol

- Atmospheric particles and coastal environments.
- Reflectance values in the deep blue part of the spectrum, which is sensitive to particles in the air.
- High values can indicate atmospheric haze or the presence of a smoke plume.
- Low values suggest a clearer atmosphere.

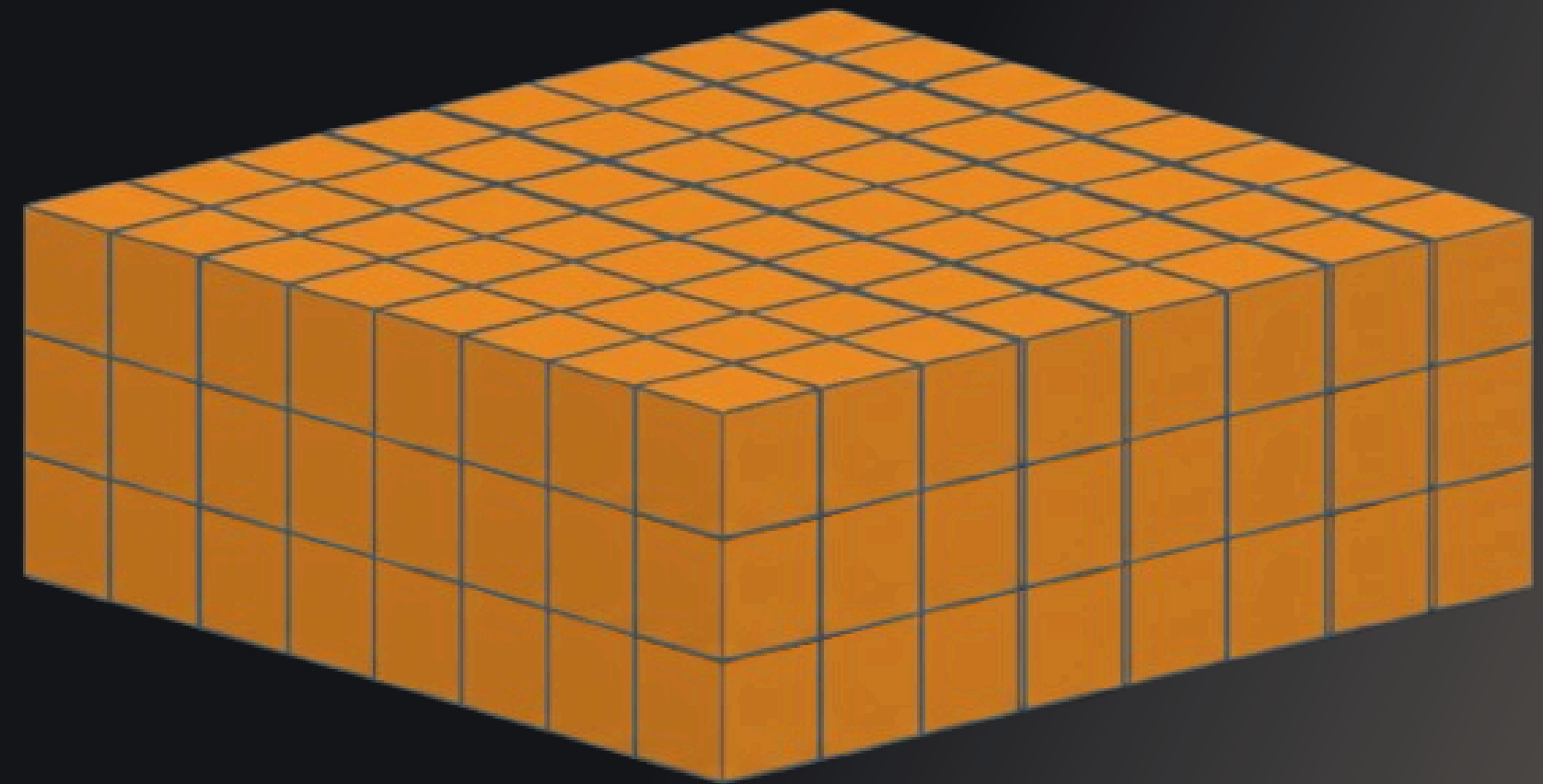




Anatomy of a Data Patch

Layer 2, 3, 4 - Visible Light (RGB)

- Capture light just as the human eye would see it.
- Reflectance values for blue (B2), green (B3) and red (B4) light.
- Dark, brownish or black areas often indicate burned ground (low reflectance).
- Bright white/orange spots can be the glow of active flames.



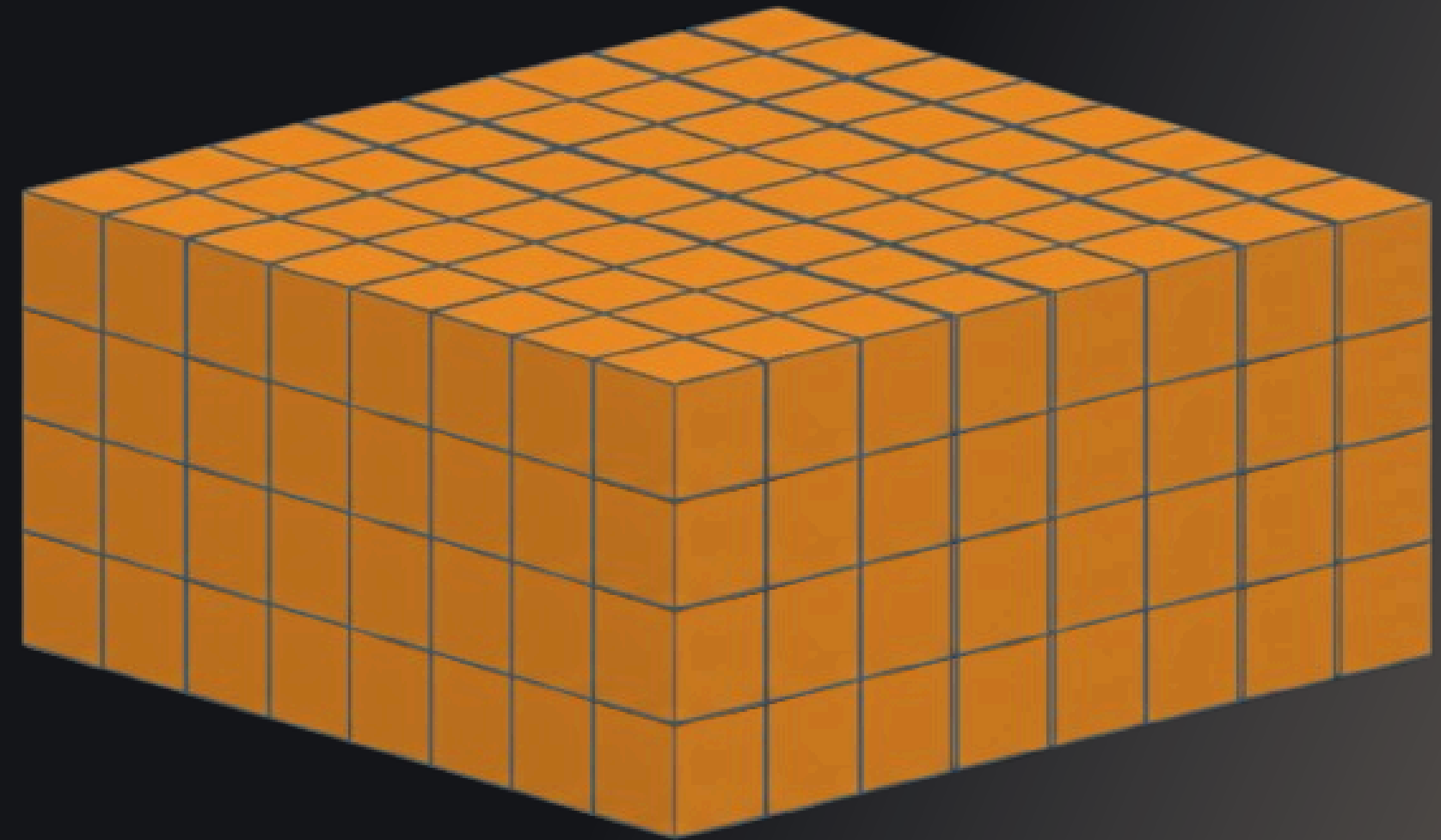


Dataset

Anatomy of a Data Patch

Layer 5, 6, 7, 9 - Vegetation Red Edge

- Highly sensitive indicators of plant health and stress.
- Reflectance values that precisely measure the health of plant chlorophyll.
- High values signify healthy, dense vegetation with active chlorophyll.
- A sharp drop in values is a key indicator that vegetation is dying or has been completely burned by fire.

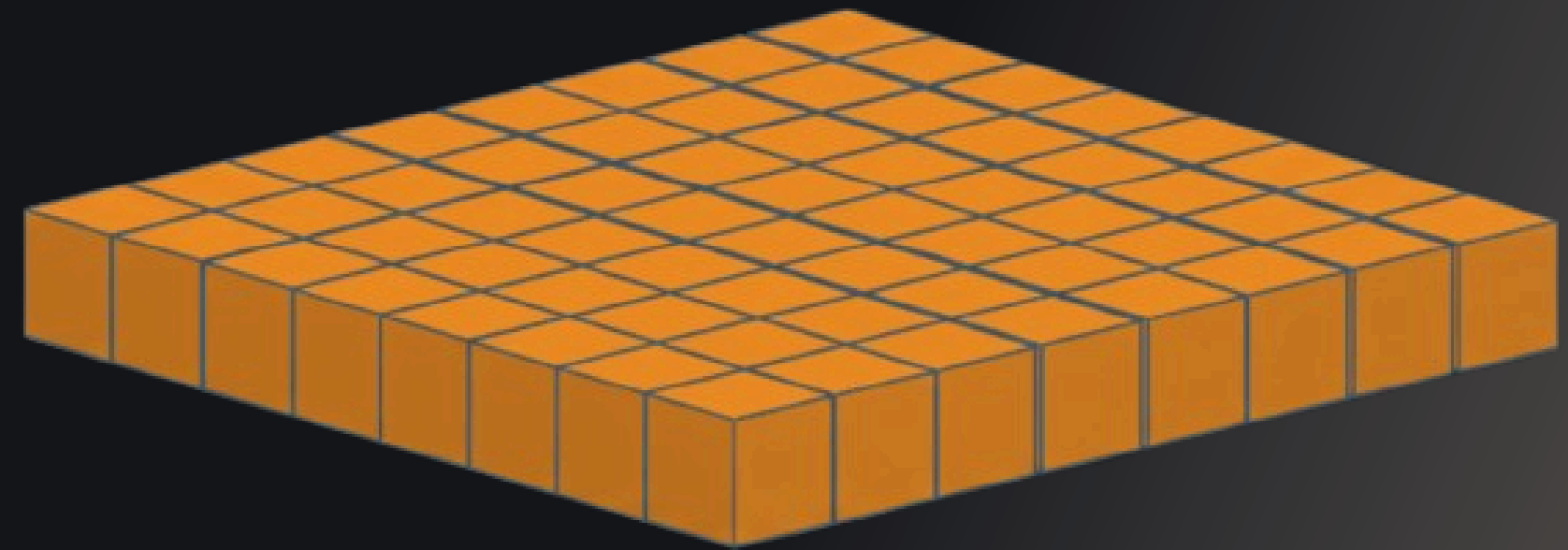




Anatomy of a Data Patch

Layer 8 - Near-Infrared (NIR)

- Distinguishing between vegetation, soil and water.
- Reflectance values in the near-infrared spectrum, which is invisible to humans.
- High values = Healthy vegetation (which reflects NIR very strongly).
- Low values = Water or burned areas (which absorb NIR).

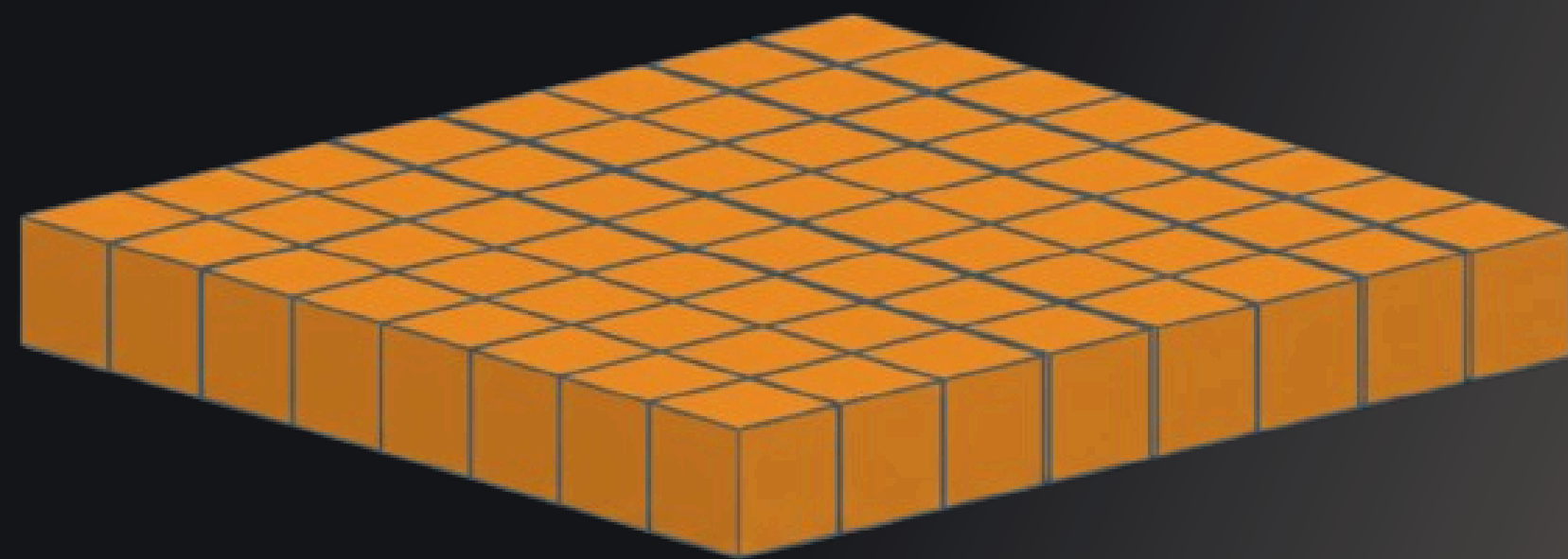




Anatomy of a Data Patch

Layer 10 - Water Vapour

- Measure the amount of water vapor in the atmosphere.
- Measures atmospheric absorption caused by water vapor between the satellite and the ground.
- While less direct for fire detection, it helps improve the accuracy of other bands. Dense smoke plumes can sometimes influence its readings.



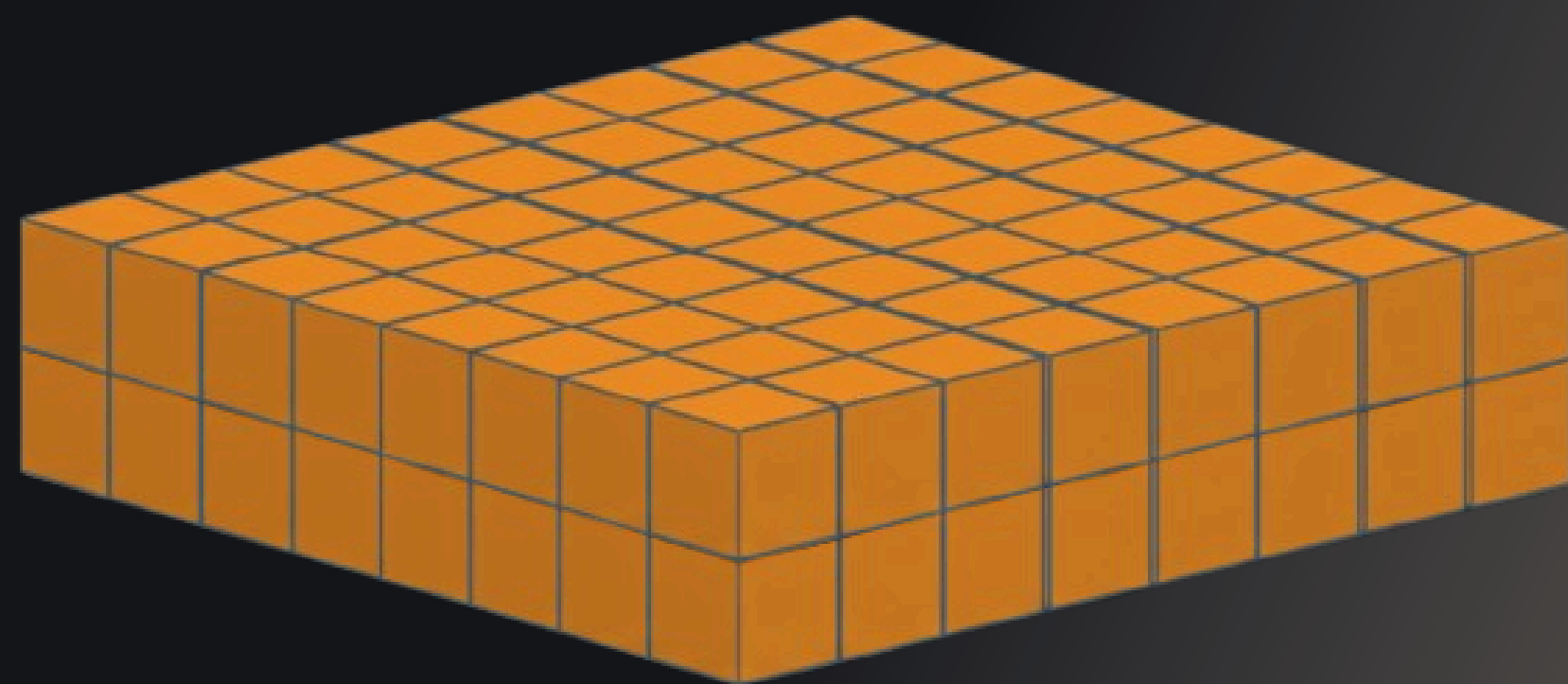


Dataset

Anatomy of a Data Patch

Layer 11, 12 - Short-Wave Infrared (SWIR)

- Highly sensitive to both temperature and moisture.
- Reflectance and thermal emission data sensitive to heat and water content.
- Very high (saturated) values are a direct signal of intense heat from active flames.
- Low values can indicate high moisture in soil/vegetation, while dry, burned areas have higher SWIR reflectance.

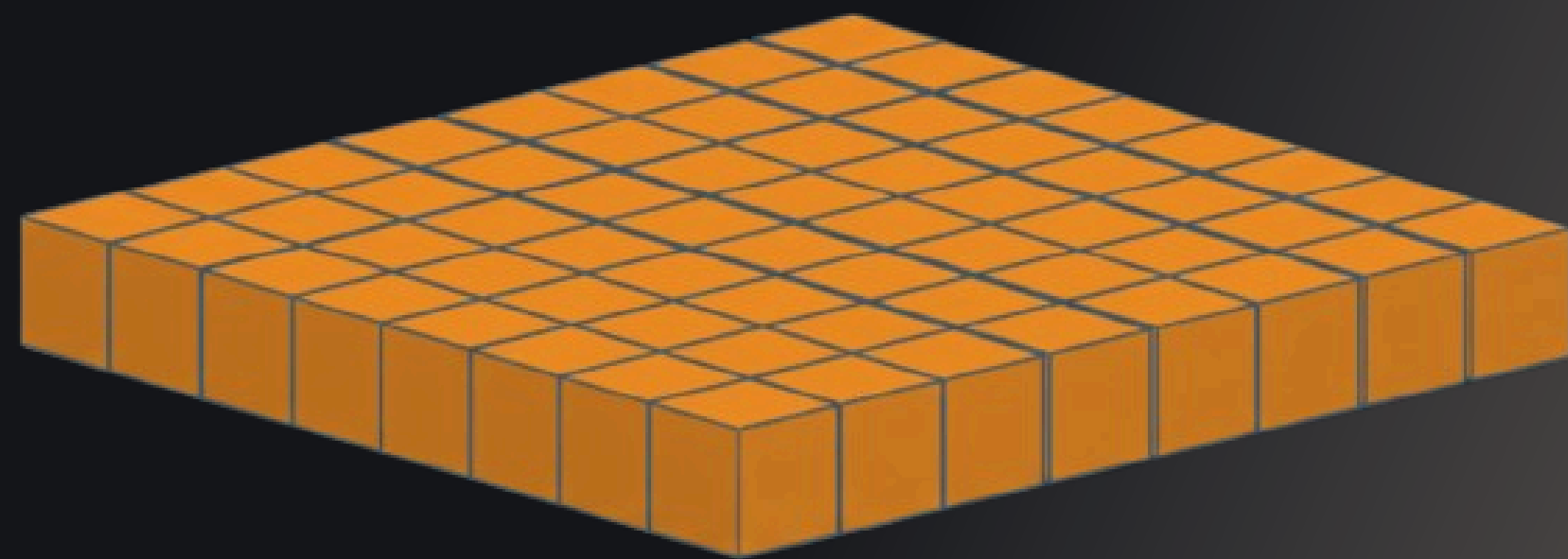




Anatomy of a Data Patch

Layer 13 - Aerosol Index

- From a different satellite (Sentinel-5P) specialized in air quality monitoring. It allows the model to detect the smoke plume directly, not just the fire on the ground.
- Indicate the density of UV-absorbing aerosols like smoke, dust, or volcanic ash.
- High values strongly indicate the presence and thickness of a smoke plume.
- Low values indicate clear air.

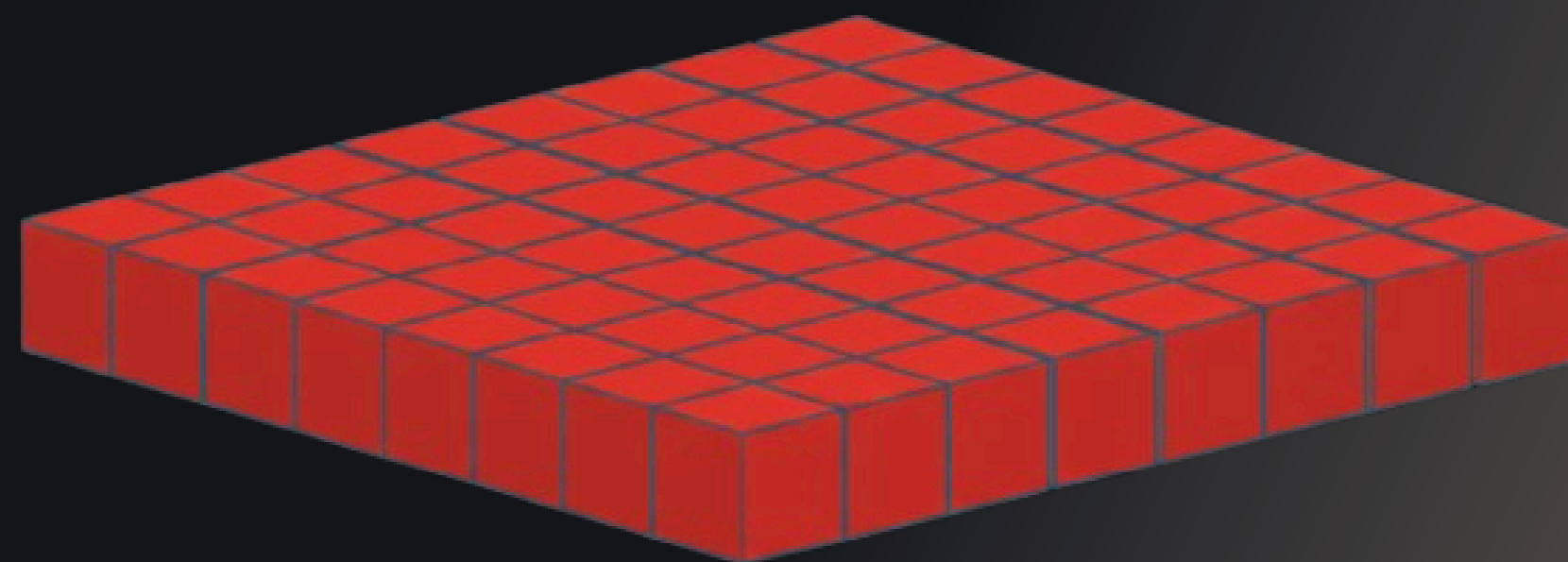




Anatomy of a Data Patch

Label Mask (Ground Truth)

- Used to evaluate the model. It tells the algorithm exactly which pixels represent a fire.
- A grid of pixel values, containing either a 0 or a 1.
- Value of 1 = This pixel is classified as 'fire'.
- Value of 0 = This pixel is classified as 'non-fire'.





Preprocessing

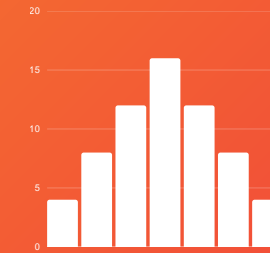
Preprocessing

Channel Selection

- Sen2Fire → 13 channels (12 Sentinel-2 + 1 Sentinel-5P).
- Better to select specific band combination rather than using all the channels.
- We will test:
 - RGB (3 channels).
 - RGB + NIR (4 channels).
 - All 13 channels.
 - Other combinations of channels.

Standardization

$$x' = (x - \mu) / \sigma$$



Apply Z-score standardization per channel.

Handle the vastly different scales of input channels.

Ensure all features contribute equally during training.

Improve model training speed and stability.



Augmentation

Augmentation

- Online augmentation of only the training set.
- Apply geometric transformation also to the label mask.

1458 $\Rightarrow$ 4374^(maybe)

Train Patches



Horizontal Flip (Left to Right)



Vertical Flip (Up to Down)



Zoom In





Metrics

METRICS



Precision (P)

$$P = \frac{TP}{TP + FP}$$

Recall (R)

$$R = \frac{TP}{TP + FN}$$

F1 Score

$$F1 = 2 \times \frac{P \times R}{P + R}$$

IoU (Intersection over Union)

$$IoU = \frac{TP}{TP + FP + FN}$$



Loss

LOSS



Binary Cross-Entropy (BCE)

Evaluates pixel-wise classification error.

Dice Loss

Focuses on overlapping areas, useful for imbalanced datasets.



Combined Loss (BCE + Dice Loss)

Balances pixel accuracy and region overlap



Models

MODELS



Baseline U-NET (CNN)

Replicating the paper's
foundation.



nnU-NET

A robust framework likely
to achieve state-of-the-art
results.



SegFormer

Testing a modern,
Transformer-only
approach.



Mixed Transformer- CNN

Combining CNN local
features with Transformer
global context.



Dual-Stream Fusion

A custom architecture
prioritizing key spectral
bands.





Training

TRAINING



On-Premise Workstation

Dual NVIDIA RTX 3060 Ti
GPUs (16GB)
16-Core CPU
128GB RAM



Cloud Platform (Kaggle)

Access to A100 NVIDIA
GPUs
Facilitates reproducible
experiments.




$$>$$


MODEL	PRECISION	RECALL	F1-SCORE
U-NET	39.7	21.8	28.1
U-NET	42.10	14.51	21.58
nnU-NET			
SegFormer			
Mixed Transformer-CNN			
Dual-Stream Fusion			





Reproducibility and Availablity

Reproducibility and Availability

Dataset

- Sen2Fire dataset open access.
 - Source: European Space Agency (ESA).
 - Sentinel-2 and Sentinel-5P satellite imagery.
 - Includes annotated fire masks.
- Publicly available for research and benchmarking.



Code & Models

- Implementation based on:
 - PyTorch framework.
 - Segmentation models library (for U-Net, nnU-Net, SegFormer).
- Code repository: GitHub (to be released after validation).
- Random seeds fixed for reproducibility.



Experimental Setup

- Standardized preprocessing (Z-score normalization).
- Consistent train/val/test splits across all models.
- Results logged and saved for verification.





Fallback Strategies

Fallback Strategies



Apply multiple augmentation techniques (flips, rotations, zooms) to increase data diversity.

Experiment with different model architectures (U-Net, nnU-Net, SegFormer, ...).

Exclude underperforming models from final presentation but report their outcomes.

Adjust hyperparameters (learning rate, batch size) if results are unstable.

Maintain robust training pipeline to handle potential dataset or GPU issues.



Limitations

Limitations

Data quality & variability

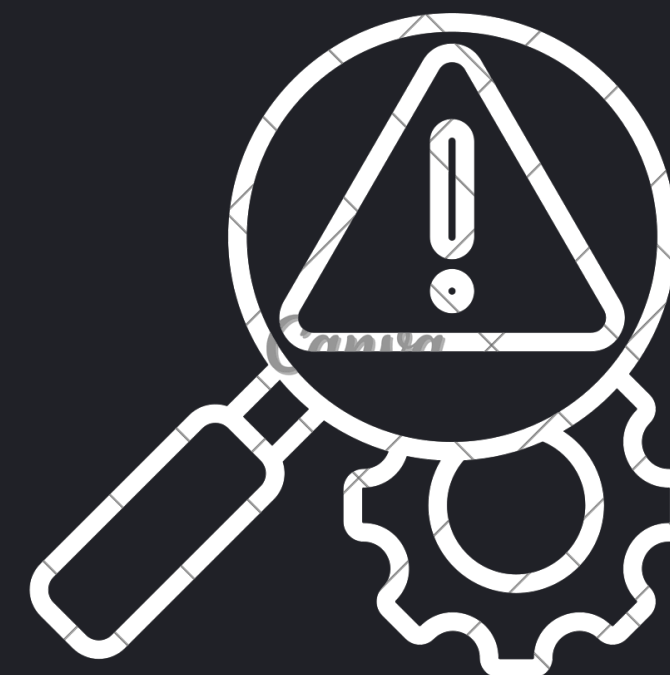
Lighting conditions and viewing angles make patterns inconsistent. Combining multiple spectral layers can introduce misleading artifacts.

Imprecise annotations

Fire and smoke boundaries are often subjective or noisy.

Generalization issues

The model may overfit to specific regions or vegetation types and perform poorly on unseen areas.





What we expect

What we expect

Improve
baseline
U-Net
results



Compare
multiple
architectures



Identify the
best-
performing
model



Gain
experience
with
various DL
frameworks



Develop a
robust
evaluation
workflow



thank you



Tito Nicola Drugman



Vlad



Kalizhan Kauymbayev

